Moodscape

From Official Statistics to Sound:

Testing Each Link of a Data-to-Music System for South Korean Places

[Participant Name] · [School Name] · Republic of Korea

KSEF 2027 International · Category [Code]

Maps are read only with the eyes, and one number is not a place

Maps depend on vision

Which regions are busy or quiet is shown only by color and labels, and does not reach people who cannot see the screen.

Sound is another channel

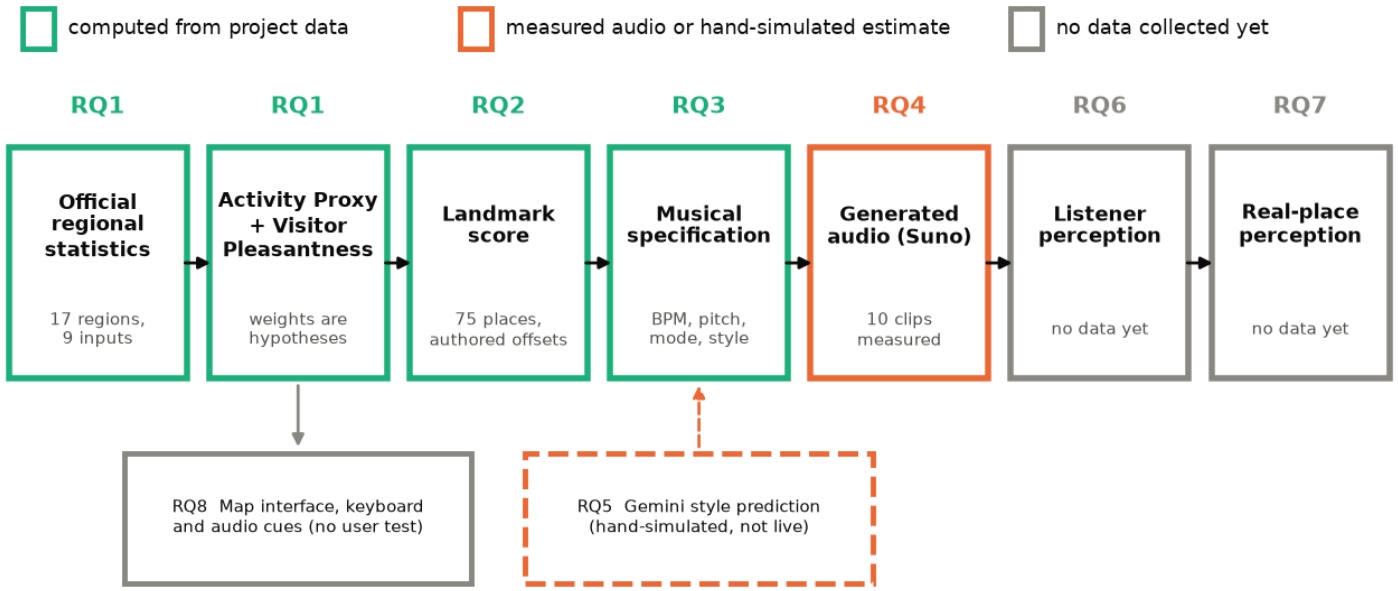
Sonified maps have helped users with visual impairment explore geographic data [Zhao 2008].

But atmosphere is perceived

A soundscape is the acoustic environment as perceived (ISO 12913); perceived sources predict perception better than sound level alone [Oberman 2025].

Question: Can official statistics be turned into music, and does each link of that chain actually work?

Goal: build it, then test every link on its own



Objectives

① Build a reproducible data-to-music web app

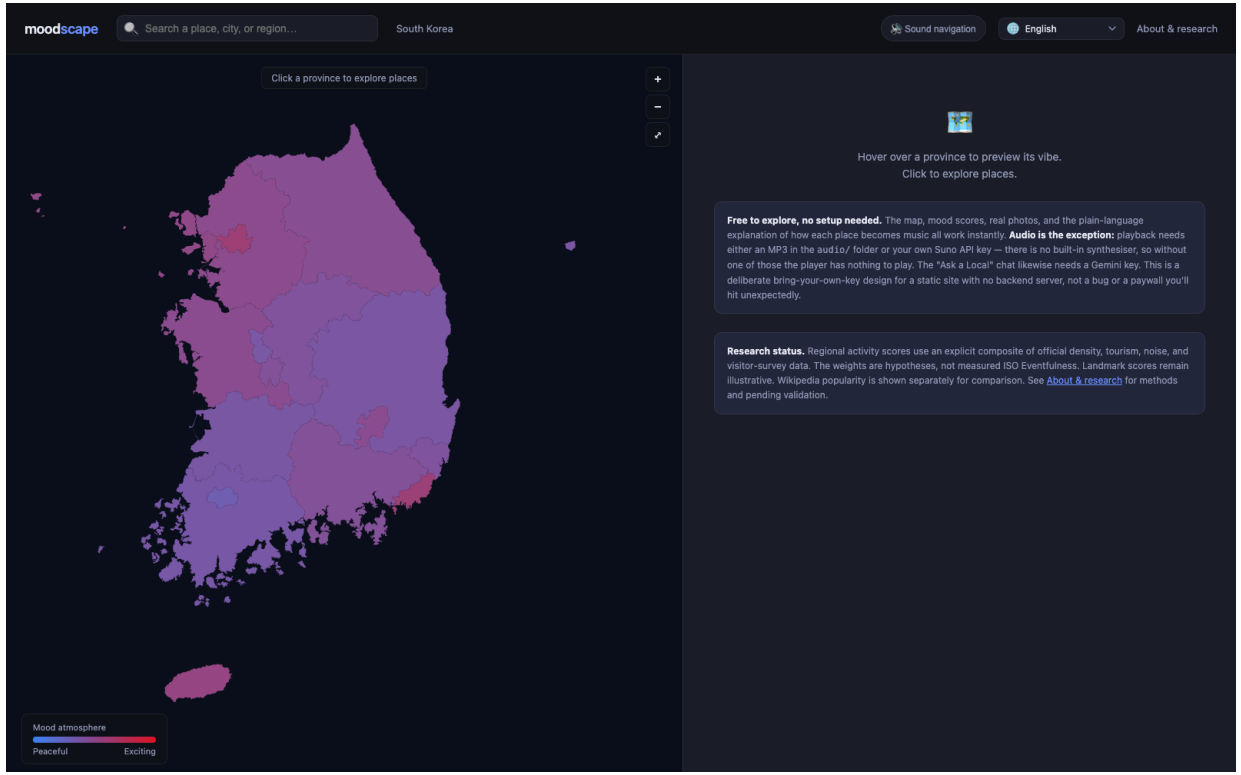
② Test each link as its own research question

③ Derive improvements from the results

Evidence classes

- Computed: RQ1-RQ3
- Measured / simulated: RQ4-RQ5
- Pending: RQ6-RQ8 (listeners, field, access)

Idea → system: activity becomes tempo, satisfaction becomes mode



Input: 9 official indicators, 17 provinces

Density (Statistics Korea), tourism and visitor survey (2024 National Travel Survey), noise complaints and facilities (Ministry of Environment), transcribed with source links.

Activity Proxy

100 × (0.35 density + 0.30 tourism + 0.15 complaints + 0.10 facilities + 0.10 crowding)

Musical specification

Tempo = 42 + 110 × score BPM · pitch = A4 + 12 × score semitones · mode = 3 rules → generated with Suno V4_5

Accessibility

Keyboard navigation, screen-reader labels, score → pitch, east-west → stereo

Design: generate the same places twice, measure by software and by ear

Variables

- Independent: wording, request shape (single text / custom), genre words
- Dependent: detected mode and tempo
- Controlled: model V4_5, analysis procedure, seed 20260920

Measurement

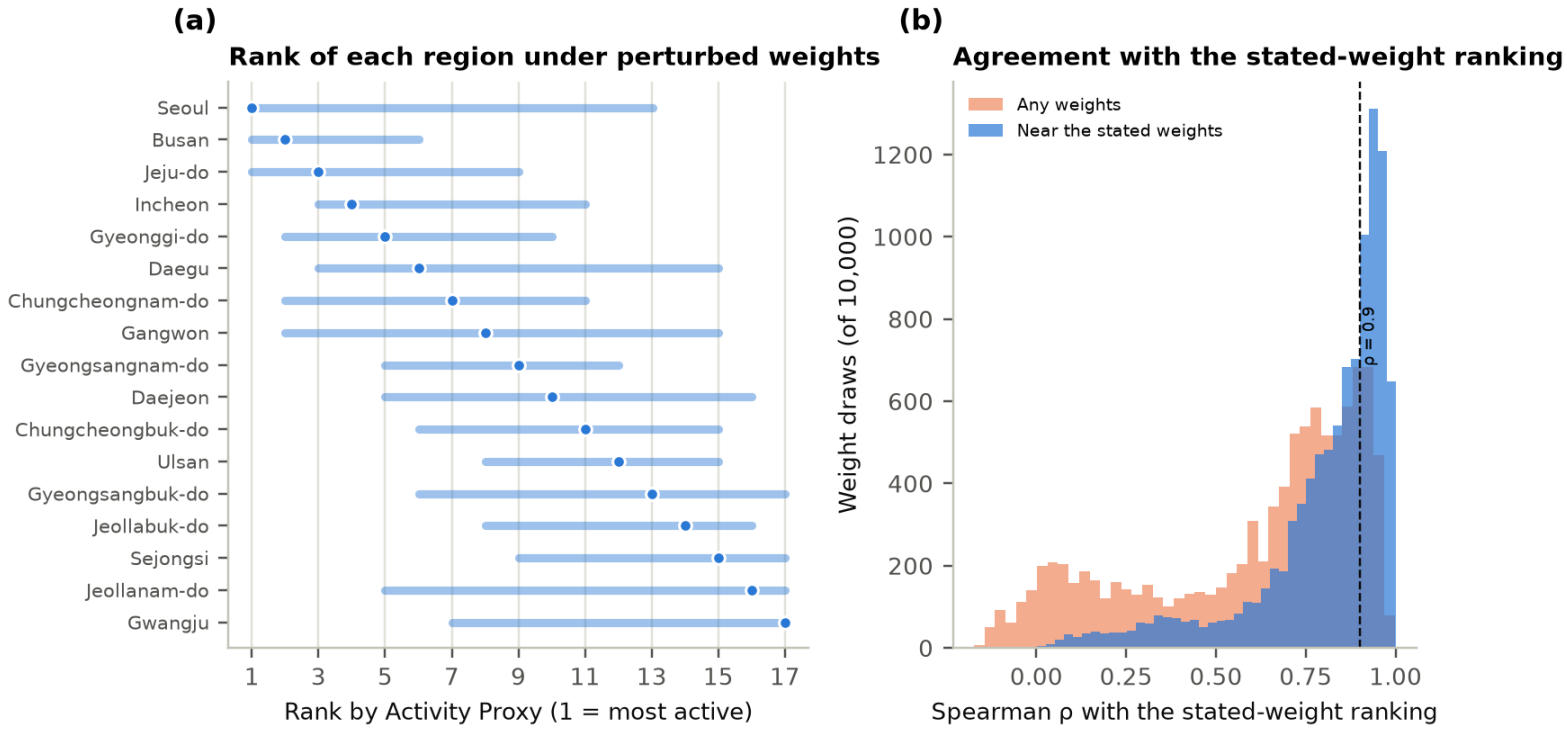
- 10 places (score quantiles) × 2 rounds
- Key: Krumhansl-Kessler profile correlation
- Tempo: single estimate + top-3 tempogram peaks, 20,000 permutations

Statistics

- Weight sensitivity: 10,000 Dirichlet draws
- Clopper-Pearson intervals, Fisher exact test
- A single success is never counted as a fix (stochastic)

Human check: one listener blind-tapped one Sejong clip (engineering diagnostic, n = 1, not the listener study)

RQ1 · The regional order is plausible but weight-sensitive



ρ = 0.87

median rank agreement under perturbed weights

42%

of draws kept ρ ≥ 0.9

ρ = -0.07

actual influence of tourism (weight 0.30)

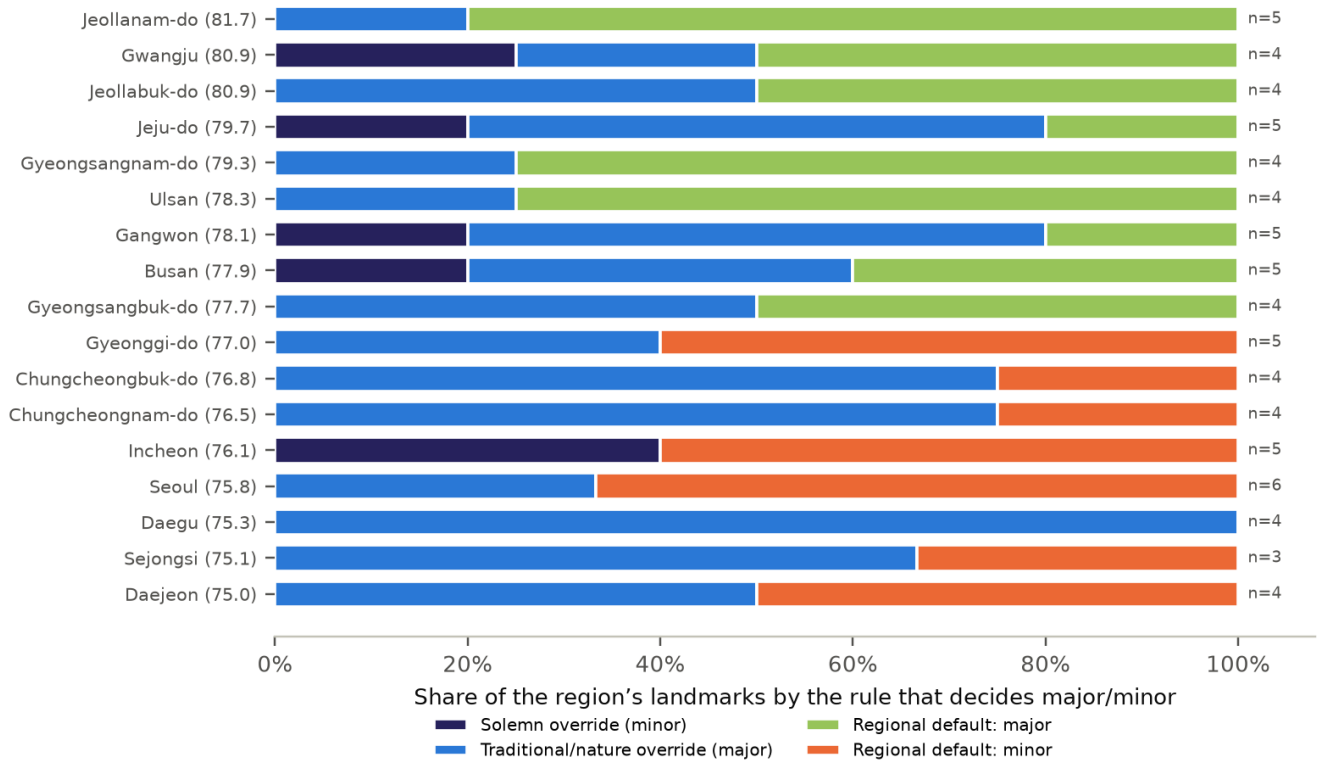
RQ2-RQ3 · Landmark sound is mostly authored; mode is set by region

12.2%

of landmark-score variance explained by region (η^2)

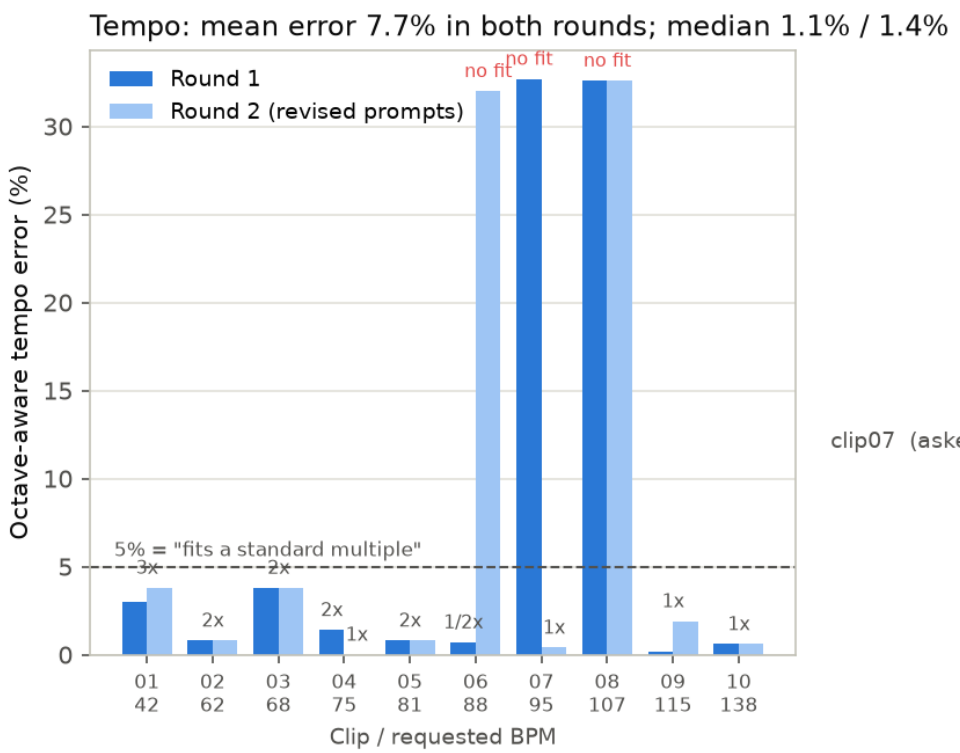
35/75

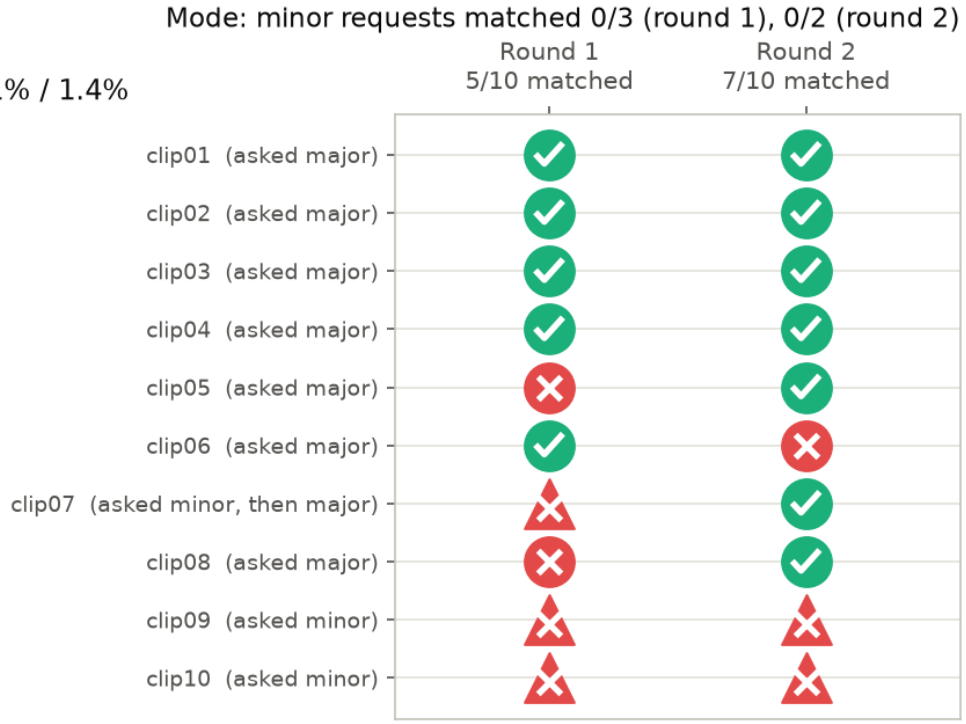
places whose mode is set by Visitor Pleasantness



The 17 Pleasantness values span 6.7 points and 6 regions sit within one point of the median, so small survey differences become major vs minor.

RQ4 · Generated audio follows the request only in part





5→7/10

mode matched (round 1→2)

0/5

minor requests honored

7·9/10

request among top-3 peaks (chance ≈1.5)

The round-2 gain mixes Sejong's changed request (minor → major) with clips 05, 06 and 08 flipping under the same request, so it cannot be credited to rewording.

An automatic 95 BPM match was heard at 48 BPM

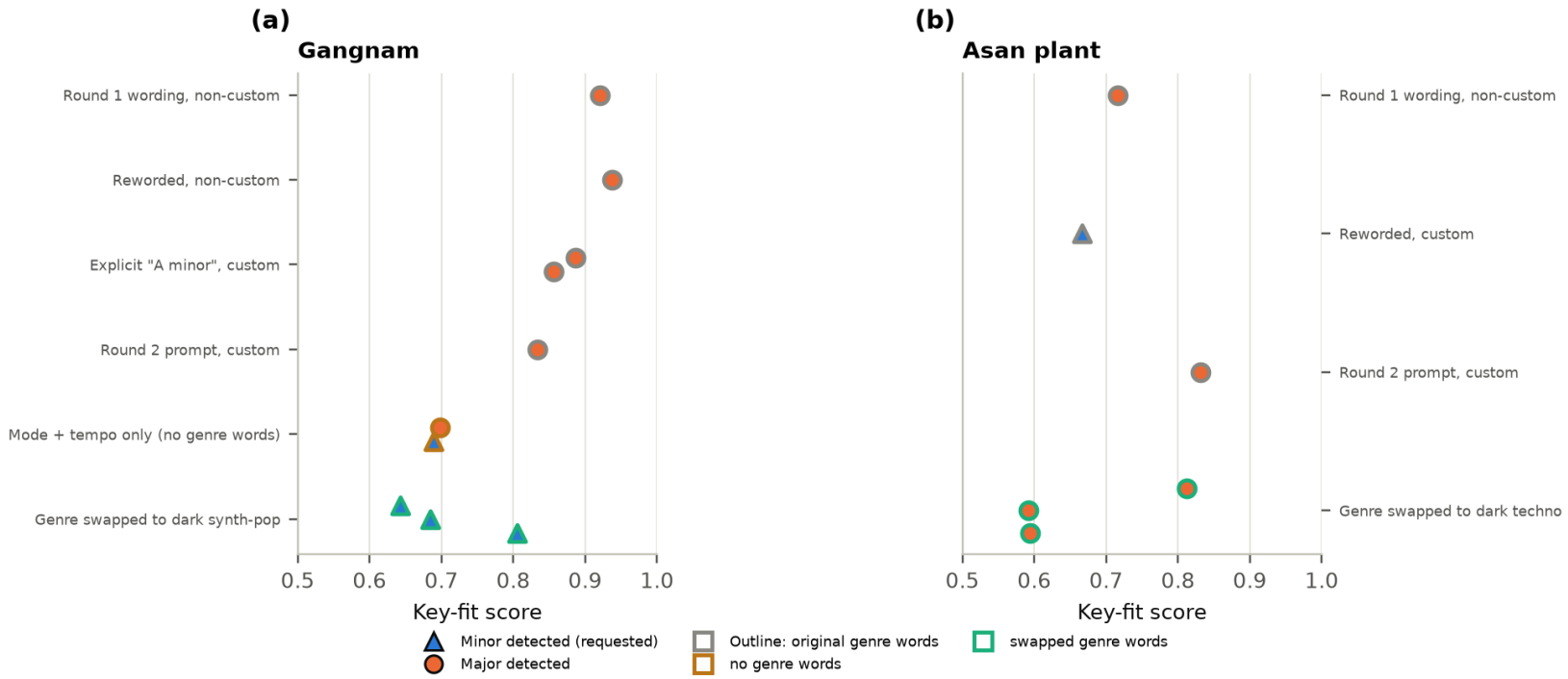
Target 95 BPM | automatic → human blind tap

Round-2 Suno source	Strict 95 BPM regeneration	+ 95 BPM kick and rim	2.02× time-axis fix + hi-hat
auto 95.3	auto 95.3	auto 95.3	auto 95.3
48 BPM	**48 BPM**	**47 BPM**	**96 BPM**

The added pulse was absorbed as a subdivision of the slow beat. Only correcting the whole time axis from the human measurement (47) moved the heard pulse to the target (1.1% error; criterion ±5% set in advance).

Limit: one place, one listener, sequential diagnostic, not a success rate. Listen to the audio

What changes the mode? Genre words more than mode words



0/5 → 3/3

Gangnam: original genre → “dark synth-pop”

0/3

Asan: same method, did not transfer

9/21

all minor requests (no factor beats chance)

Feedback → fixes: the biggest gains came from fixing our own rules

“complex” keyword collision → Sejong styled as a factory	→	**New institutional category**	later generations at 95.3 BPM
Sejong is a planned city of young families (user feedback)	→	**Added to the major rule**	D♯ major, key-fit 0.94
“warm, mellow” next to a minor request	→	**Remove bright words when minor**	necessary, not sufficient
Automatic BPM ≠ human pulse	→	**Time-axis fix + hi-hat**	one listener: 96 BPM
API key routed through a public relay	→	**Direct calls**	checked with a mocked network

Limitations and next steps

Limitations

- 20 tracks + ~20 follow-ups, one provider and model
- Follow-up comparisons were not preregistered
- Only one listener checked one clip
- Weights are hypotheses; landmark scores are not measured
- No listener, field or accessibility data yet (RQ6-8)
- No recorded Gemini output (hand-simulated)

Next steps

1. Several listeners tap every clip blind; a musician checks mode
2. Repeat each request before claiming a fix
3. Test the genre-word hypothesis at more places
4. Listener study, 15-24 adults (protocol ready)
5. Field soundwalk at 6-10 sites
6. Enforce tempo by deterministic correction

Conclusion

- The chain from official statistics to score, specification and audio is traceable end to end, and its first four links were tested.
- The regional order is plausible but weight-sensitive, landmark scores are mostly authored, and the mode rule is coarse.
- The generator guaranteed neither minor mode (0/5) nor the perceived pulse; deterministic correction fixed tempo (one listener, one clip).
- Whether listeners hear the intended order is the next, prepared experiment.

Materials · Code & data · Web demo · Generated audio

https://github.com/clairenah19/sound-moodscape/tree/ksef2027-submission